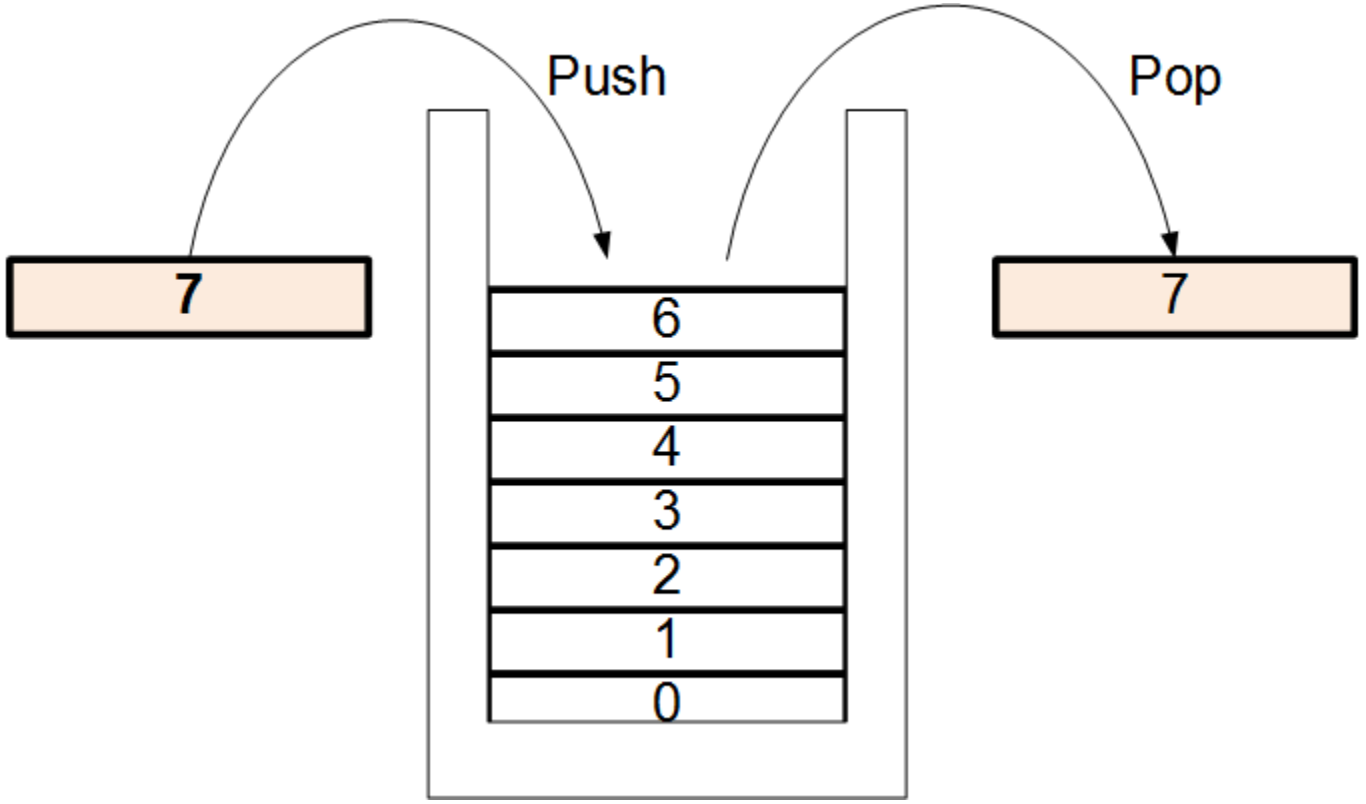
Introduction

LIFO (Last-In, First-Out), also known as a **Stack**, is an important data structure used to store and retrieve data in the reverse order of its entry: the data entered last will be retrieved first.

LIFO has two important parameters: the number of memory cells and the data width, which determine the LIFO capacity. Two characteristic parameters of LIFO are:

- The number of memory cells, also known as LIFO depth.
- Memory cell width, corresponding to the width of the data being written and read.

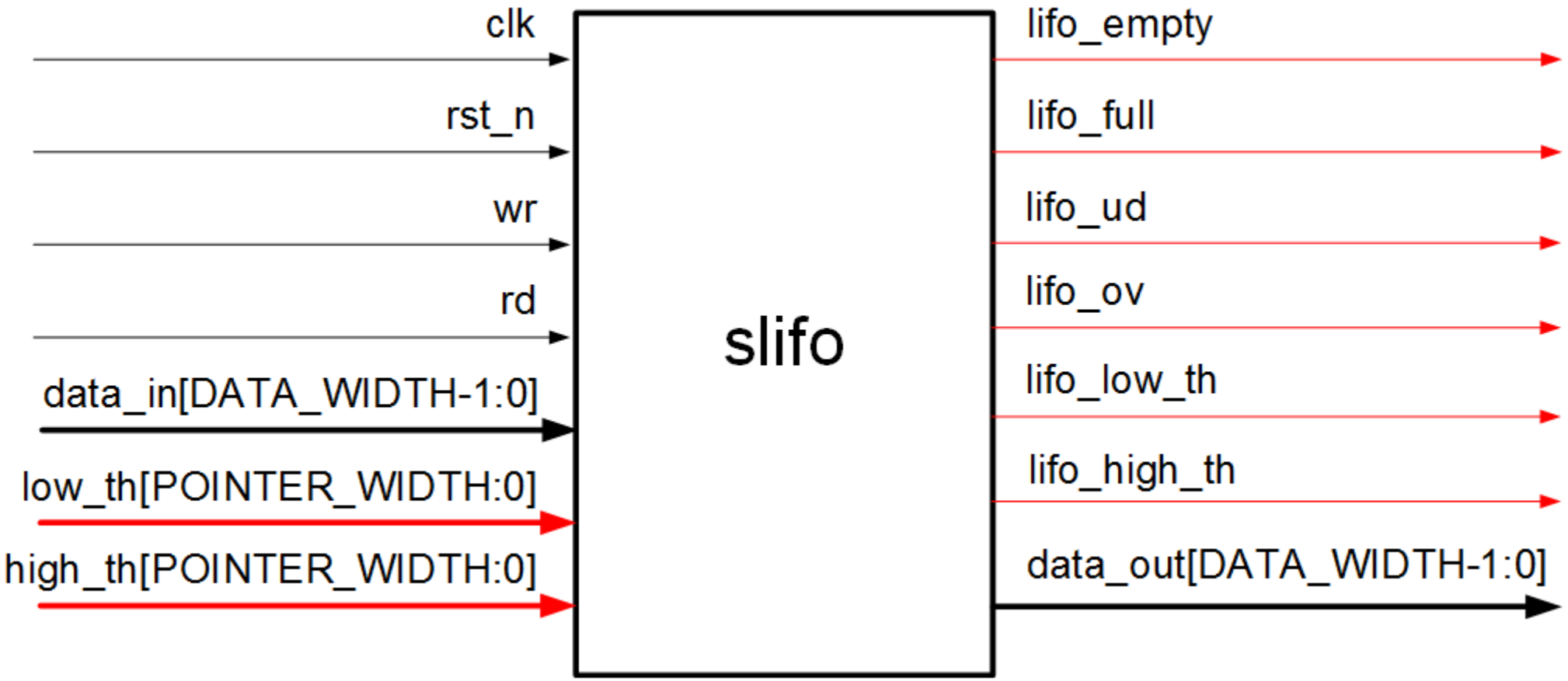
The two parameters above will determine the LIFO capacity. For example, if a LIFO has 4 memory cells and each cell stores 8 bits of data, the LIFO capacity is 4 * 8 = 32 bits or 4 bytes. To address the memory cells of the LIFO, a commonly used method is to use a write pointer and a read pointer. The pointer here is essentially a sequential counter.



LIFO usually has signals to report its operating status, such as:

Status	Description
Empty State	Does not contain any data.
Full State	The entire capacity has been filled.
Overflow State	It is full but writing continues. In this case, the data in LIFO remains unchanged and is not overwritten by new data.
Underflow State	It is empty but reading continues. In this case, the data output will hold the most recent old data value.
Threshold State (below or above a threshold value)	For example, if the amount of data in LIFO is less than or equal to 4, the low threshold is activated. The value 4 here is the low threshold.

HIGH LEVEL BLOCK DESIGN



Input:

Port	Description
clk	Clock signal.
rst_n	Active-low reset signal.
wr	Data write signal into LIFO.
rd	Data read signal from LIFO.
data_in[DATA_WIDTH-1:0]	Data write bus with DATA_WIDTH bits.
low_th[POINTER_WIDTH-1:0]	Low threshold configuration signal with POINTER_WIDTH bits.
high_th[POINTER_WIDTH-1:0]	High threshold configuration signal with POINTER_WIDTH bits.

Output:

Port	Description
lifo_empty	Signal reporting LIFO is empty.
lifo_full	Signal reporting LIFO is full.
lifo_low_th	Signal reporting LIFO is below the low threshold.
lifo_high_th	Signal reporting LIFO is higher than or equal to the high threshold.
lifo_ov	Signal reporting LIFO overflow.
lifo_ud	Signal reporting LIFO underflow.
lifo_data_out[DATA_WIDTH-1:0]	Data read bus with DATA_WIDTH bits.

Parameters:

- DATA_WIDTH: Sets the number of bits for 1 LIFO data cell, corresponding to the width of the write/read data bus.
- POINTER_WIDTH: Sets the width of the write/read pointer. This value will determine the number of LIFO memory cells according to the formula: Number of LIFO memory cells =2^POINTER_WIDTH. For example, if POINTER_WIDTH is configured as 3, the number of LIFO memory cells is 2^3 = 8 cells.

Note: This LIFO does not allow simultaneous reading and writing, meaning only reading or writing can occur at a single point in time.

Read/Write Pointer

The read and write pointer for this LIFO design is just a single counter that operates according to the principle:

- Increases by 1 when writing.
- Decreases by 1 when reading.

The read/write pointer circuit can be implemented in two ways:

- Using 1 adder and 1 subtractor.
- Using 1 adder and 1 MUX. This method is based on the principle of converting a subtractor into a 2's complement addition circuit.

